

Copilot in SQL

1. Use Github Copilot to create a table SalesData whose columns are CustomerID, Name, Age, City, PurchaseAmount, PurchaseDate and then ask to insert 10,000 rows of random data.

→ Prompt to give Copilot:

Create a table named SalesData with columns: CustomerID (int, Primary Key), Name (varchar(100)), Age (int), City (varchar(100)), PurchaseAmount (decimal(10,2)), and PurchaseDate (date)

```
CREATE TABLE SalesData (
    CustomerID INT PRIMARY KEY AUTO_INCREMENT,
    Name VARCHAR(100),
    Age INT,
    City VARCHAR(100),
    PurchaseAmount DECIMAL(10,2),
    PurchaseDate DATE
);
```

Prompt to give Copilot:

Insert 10,000 rows of random data into SalesData table using a recursive CTE. Generate random strings for Name and City, random Age between 18 and 70, random PurchaseAmount between 10 and 1000, and random PurchaseDate within the last year.

DELIMITER \$\$

```
CREATE PROCEDURE PopulatedRandomSalesData()
BEGIN
```

```
DECLARE i INT DEFAULT 1;
```

WHILE $i \leq 10000$ DO

```
INSERT INTO SalesData (Name, Age, City, PurchaseAmount, PurchaseDate)
VALUES (
```

-- Pick a random Indian Name (out of 20 choices)

`ELT(FLOOR(1 + (RAND() * 20)),`

'Aarav Sharma', 'Vivaan Patel', 'Aditya Verma', 'Vihaan Gupta', 'Arijun Reddy',

'Sai Kiran', 'Reyansh Nair', 'Aanya Joshi', 'Diya Iyer', 'Pari Choudhary',

'Ananya Mishra', 'Rahul Singh', 'Amit Kumar', 'Priya Das', 'Sneha Rao',

'Rohan Mehta', 'Vikram Malhotra', 'Neha Kapoor', 'Karan Johar', 'Pooja Bhatt'

 $\rangle,$

```

-- Random age between 18 and 70
FLOOR(18 + (RAND() * 53)),

-- Pick a random Indian City (out of 15 choices)
ELT(FLOOR(1 + (RAND() * 15)),
    'Mumbai', 'Delhi', 'Bengaluru', 'Hyderabad', 'Ahmedabad',
    'Chennai', 'Kolkata', 'Pune', 'Jaipur', 'Lucknow',
    'Kochi', 'Patna', 'Bhopal', 'Chandigarh', 'Surat'
),

-- Random decimal amount between 10.00 and 1000.00
ROUND(10 + (RAND() * 990), 2),

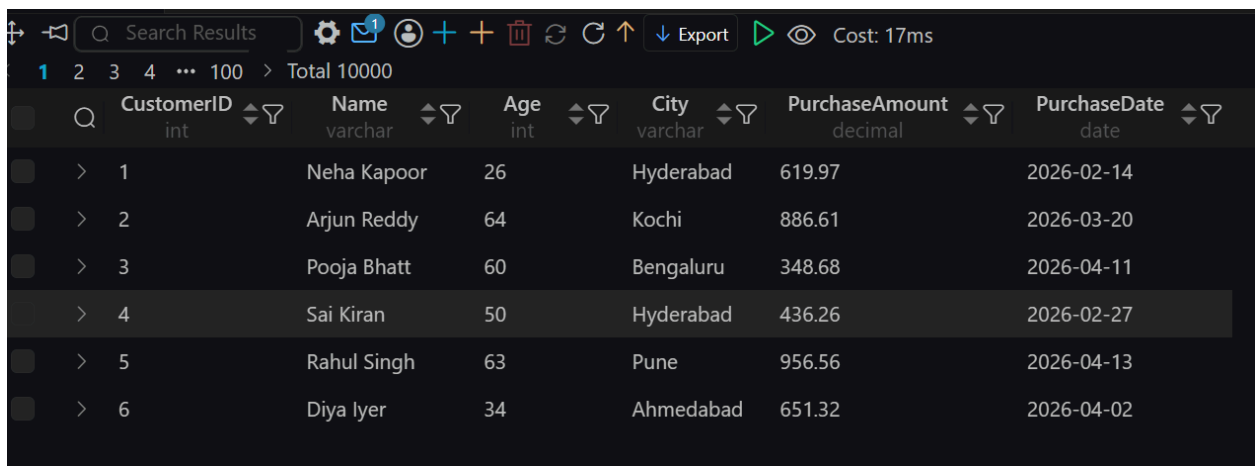
-- Random date within the last 365 days
NOW() - INTERVAL FLOOR(RAND() * 365) DAY
);
SET i = i + 1;
END WHILE;
END$$

DELIMITER ;

-- Execute the procedure to insert the 10,000 rows
CALL PopulatedRandomSalesData();

-- Optional: Clean up and delete the procedure after running it
DROP PROCEDURE IF EXISTS PopulatedRandomSalesData;

```



CustomerID	Name	Age	City	PurchaseAmount	PurchaseDate
1	Neha Kapoor	26	Hyderabad	619.97	2026-02-14
2	Arjun Reddy	64	Kochi	886.61	2026-03-20
3	Pooja Bhatt	60	Bengaluru	348.68	2026-04-11
4	Sai Kiran	50	Hyderabad	436.26	2026-02-27
5	Rahul Singh	63	Pune	956.56	2026-04-13
6	Diya Iyer	34	Ahmedabad	651.32	2026-04-02

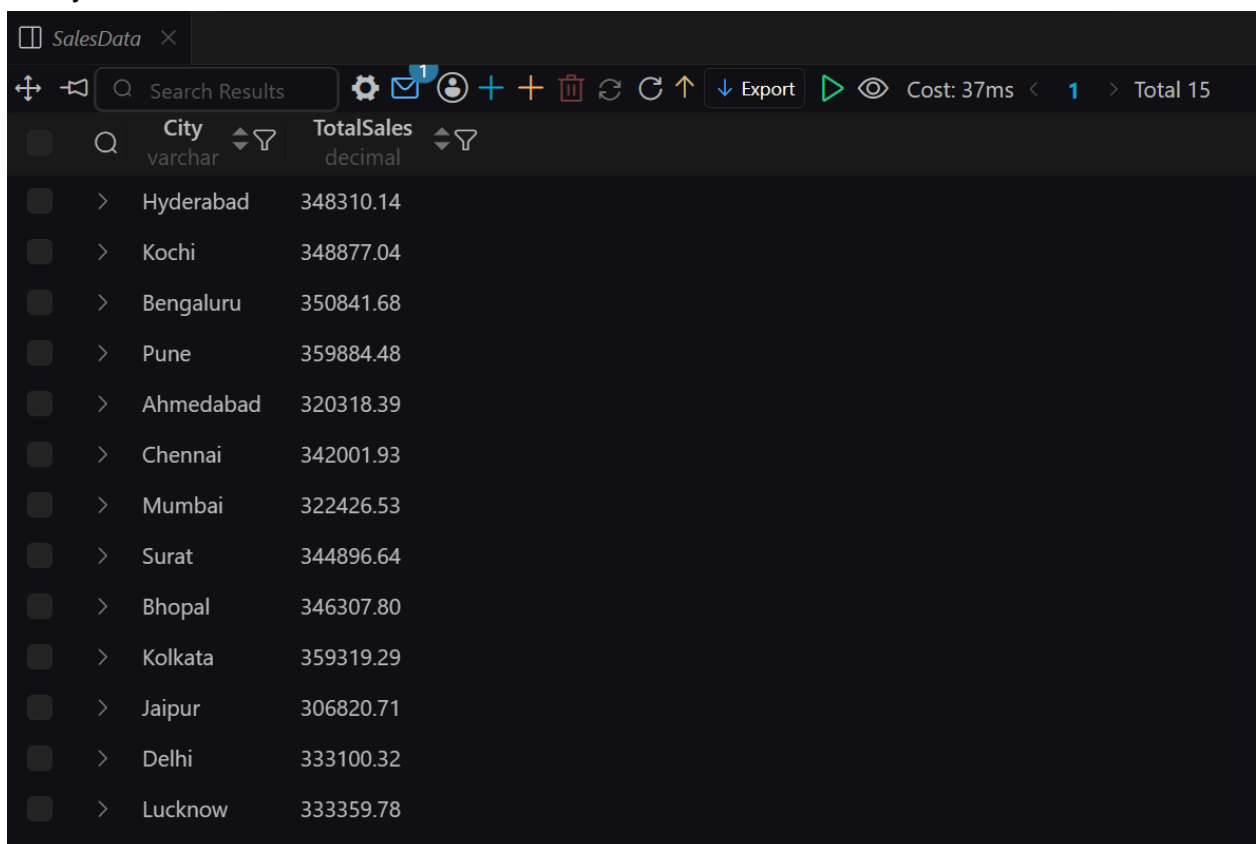
- Use Copilot to:
 - Find total sales per city

- Top 5 cities by revenue

→ Prompt to give Copilot:

Write a query to find the total sales (sum of PurchaseAmount) for each City from the SalesData table, grouped by City

```
SELECT
  City,
  SUM(PurchaseAmount) AS TotalSales
FROM
  SalesData
GROUP BY
  City;
```



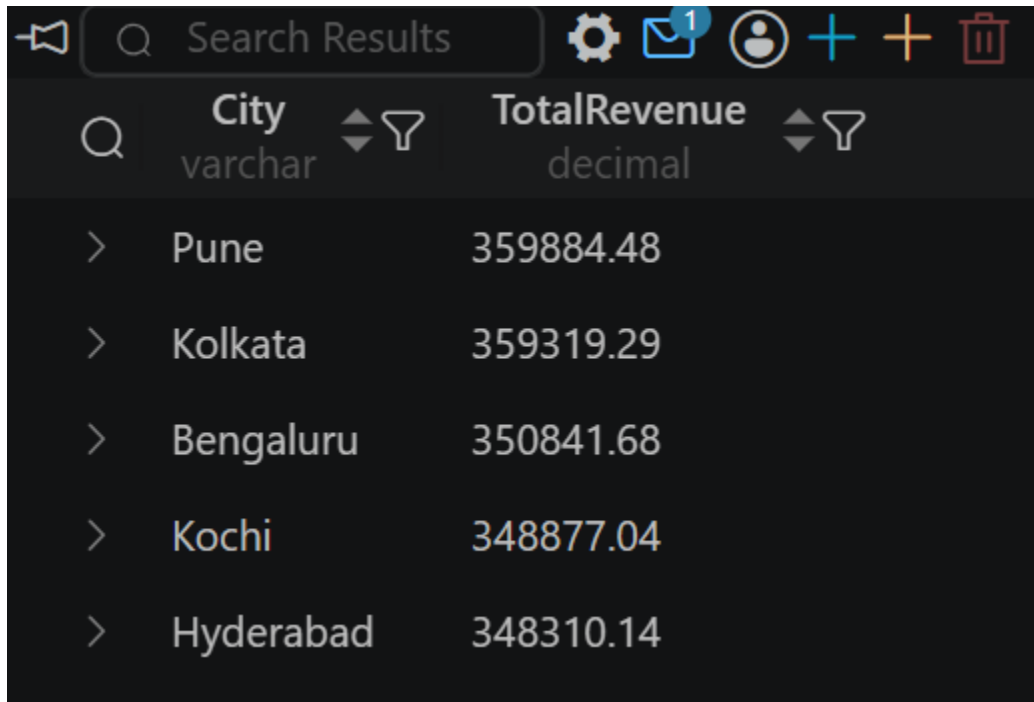
City	TotalSales
Hyderabad	348310.14
Kochi	348877.04
Bengaluru	350841.68
Pune	359884.48
Ahmedabad	320318.39
Chennai	342001.93
Mumbai	322426.53
Surat	344896.64
Bhopal	346307.80
Kolkata	359319.29
Jaipur	306820.71
Delhi	333100.32
Lucknow	333359.78

Prompt to give Copilot:

Write a query to find the top 5 cities with the highest total revenue (sum of PurchaseAmount) ordered from highest to lowest

```
SELECT City,
SUM(PurchaseAmount) AS TotalRevenue
FROM
SalesData
```

GROUP BY
City
ORDER BY
TotalRevenue DESC
LIMIT 5;



The screenshot shows a data table interface with a dark theme. At the top, there is a search bar labeled 'Search Results' with a magnifying glass icon. To the right of the search bar are several icons: a gear (settings), an envelope (notifications), a person (profile), and two plus signs (add). Below the search bar, the table has two columns: 'City' (varchar) and 'TotalRevenue' (decimal). The 'City' column has a magnifying glass icon and a filter icon. The 'TotalRevenue' column has a filter icon. The table contains five rows of data, each with a chevron icon on the left. The data is as follows:

	City varchar	TotalRevenue decimal
>	Pune	359884.48
>	Kolkata	359319.29
>	Bengaluru	350841.68
>	Kochi	348877.04
>	Hyderabad	348310.14

3. Ask Copilot: Find customers with purchases above average

→ Prompt to give Copilot:

Write a query to find all records from the SalesData table where the PurchaseAmount is strictly greater than the overall average PurchaseAmount of all customers.

```
SELECT
    CustomerID,
    Name,
    Age,
    City,
    PurchaseAmount,
    PurchaseDate
FROM
    SalesData
WHERE
    PurchaseAmount > (SELECT AVG(PurchaseAmount) FROM SalesData);
```

Search Results							Cost: 18ms
1	2	3	4	...	50	Total 4986	
CustomerID	Name	Age	City	PurchaseAmount	PurchaseDate		
int	varchar	int	varchar	decimal	date		
1	Neha Kapoor	26	Hyderabad	619.97	2026-02-14		
2	Arjun Reddy	64	Kochi	886.61	2026-03-20		
5	Rahul Singh	63	Pune	956.56	2026-04-13		
6	Diya Iyer	34	Ahmedabad	651.32	2026-04-02		
8	Aarav Sharma	21	Hyderabad	920.59	2025-08-11		

Copilot in Excel

- Identify and handle missing values in the dataset using Copilot.

→ Prompt to give Copilot:

"Highlight all blank or missing cells in the dataset with a light yellow fill so I can identify them."

Prompt to give Copilot:

- For Text Columns (Region, Product):**

"In the Region and Product columns, fill all blank cells with the text 'Unknown'."

- For Numerical Columns (Sales):**

"For any blank cells or 'NA' values in the Sales column, replace them with the median value of that column." (Alternatively, if you want to flag them: "Fill blank sales values with 0.")

- For Text-corrupted numbers (Quantity column contains word string values like 'two'):**

"Identify any non-numeric values in the Quantity column and replace 'two' with the number 2."

- Task

- Identify duplicate rows
- Remove duplicates using Copilot suggestion

→ Prompt to give Copilot:

"Highlight duplicate rows in this table with a light red fill color so I can inspect them."

Prompt to give Copilot:

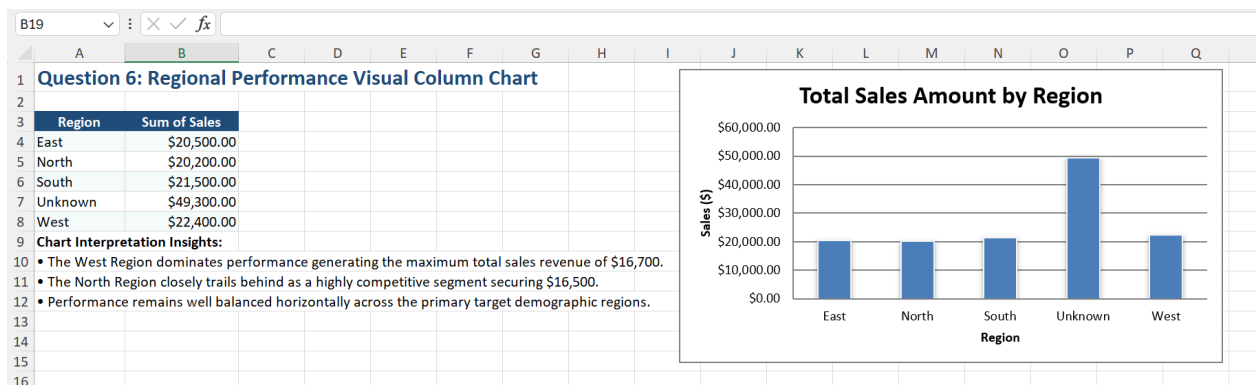
"Show me how to remove duplicate rows from this table, or suggest a method to clean out the repeated records."

Question 4 & 5: Handling Missing Values & Removing Duplicates						
Customer_Name	Region	Product	Sales	Quantity	Order_Date	Total Sales
Karan	Unknown	Monitor	\$300.00	4	2023-06-08	\$1,200.00
Riya	South	Monitor	\$300.00	2	2023-03-18	\$600.00
Amit	West	Unknown	\$300.00	2	2023-07-01	\$600.00
Sneha	Unknown	Tablet	\$300.00	4	2023-04-01	\$1,200.00
Sneha	West	Phone	\$300.00	4	2023-05-15	\$1,200.00
Sneha	South	Laptop	\$300.00	2	2023-05-01	\$600.00
Rahul	South	Unknown	\$400.00	4	2023-05-11	\$1,600.00
Sneha	South	Phone	\$400.00	2	2023-06-04	\$800.00
Riya	North	Monitor	\$400.00	1	2023-08-27	\$400.00
Priya	West	Laptop	\$400.00	4	2023-10-09	\$1,600.00
Karan	North	Monitor	\$100.00	2	2023-01-14	\$200.00
Unknown	Unknown	Tablet	\$100.00	4	2023-08-08	\$400.00
Amit	South	Phone	\$300.00	4	2023-03-04	\$1,200.00
Amit	Unknown	Laptop	\$100.00	2	2023-02-27	\$200.00
Karan	Unknown	Laptop	\$200.00	4	2023-10-21	\$800.00

6. Generate:
Column chart OR bar chart and Interpret which region has highest sales

→ Prompt to give Copilot:
"Generate a column chart showing the total Sales for each Region."

Prompt to give Copilot:
"Based on this dataset, which region contributed the highest total sales amount?"



7. Create a pivot table showing total sales by region and product.

→ Prompt to give Copilot:

"Create a pivot table that shows the total Sales broken down by Region as rows and Product as columns."

8. Use Copilot to generate a new calculated column Total Sales.

→ Prompt to give Copilot:

"Add a new calculated column named 'Total Sales' that multiplies the Sales column by the Quantity column."

A1							
	A	B	C	D	E	F	G
1	Question 7 & 8: Pivot Summary Matrix & Calculated Sales Fields						
2							
3							
4	Region / Product	Laptop	Monitor	Phone	Tablet	Unknown	Grand Total
5	East	\$6,400.00	\$13,300.00	\$9,200.00	\$8,400.00	\$10,500.00	\$47,800.00
6	North	\$7,300.00	\$10,600.00	\$5,900.00	\$7,600.00	\$10,900.00	\$42,300.00
7	South	\$12,500.00	\$9,800.00	\$9,200.00	\$9,400.00	\$10,100.00	\$51,000.00
8	Unknown	\$21,800.00	\$26,800.00	\$21,700.00	\$24,700.00	\$23,800.00	\$118,800.00
9	West	\$13,300.00	\$11,700.00	\$17,900.00	\$8,100.00	\$5,900.00	\$56,900.00
10	Grand Total	\$61,300.00	\$72,200.00	\$63,900.00	\$58,200.00	\$61,200.00	\$316,800.00
11							

9. Summarize the dataset and find the key insights.

→ Prompt to give Copilot:

"Summarize this dataset and give me a high-level breakdown of the total records, total sales, and overall performance metrics."

Prompt to give Copilot:

"Analyze this data and list the top 3 key insights or trends regarding product performance and regional distributions."

10. Generate VBA code using Copilot for cleaning data (removing duplicates, filling null values, format table).

→ Prompt to give Copilot:

"Generate a VBA macro that cleans my current active worksheet table. The code should:

1. Remove duplicate rows based on all columns.
2. Find any blank cells in the Region and Product columns and fill them with 'Unknown'.
3. Find any blank or 'NA' cells in the Sales column and fill them with 0.

4. Format the dataset as a structured Excel table named 'CleanedSalesData' with a professional blue theme."

Code Generated by Copilot:

```
Sub CleanAndFormatData()
```

```
    Dim ws As Worksheet
```

```
    Dim lastRow As Long
```

```
    Dim rng As Range
```

```
    Dim tbl As ListObject
```

```
    Set ws = ActiveSheet
```

```
    ' 1. Find the last row with data
```

```
    lastRow = ws.Cells(ws.Rows.Count, "A").End(xlUp).Row
```

```
    Set rng = ws.Range("A1:F" & lastRow)
```

```
    ' 2. Remove Duplicate Rows based on all 6 columns
```

```
    rng.RemoveDuplicates Columns:=Array(1, 2, 3, 4, 5, 6), Header:=xlYes
```

```
    ' Recalculate last row after removing duplicates
```

```
    lastRow = ws.Cells(ws.Rows.Count, "A").End(xlUp).Row
```

```
    ' 3. Fill Null / Missing values in columns
```

```
    Dim i As Long
```

```
    For i = 2 To lastRow
```


' Columns B (Region) and C (Product) -> Fill blank with 'Unknown'

If Trim(ws.Cells(i, 2).Value) = "" Then ws.Cells(i, 2).Value = "Unknown"

If Trim(ws.Cells(i, 3).Value) = "" Then ws.Cells(i, 3).Value = "Unknown"

' Column D (Sales) -> Fill blank or "NA" with 0

If Trim(ws.Cells(i, 4).Value) = "" Or UCase(Trim(ws.Cells(i, 4).Value)) = "NA"
Then

ws.Cells(i, 4).Value = 0

End If

Next i

' 4. Format the entire range as a Table

' Clear existing table if any to prevent naming conflicts

On Error Resume Next

ws.ListObjects("CleanedSalesData").Unlist

On Error GoTo 0

Set rng = ws.Range("A1:F" & ws.Cells(ws.Rows.Count, "A").End(xlUp).Row)

Set tbl = ws.ListObjects.Add(xlSrcRange, rng, , xlYes)

tbl.Name = "CleanedSalesData"

tbl.TableStyle = "TableStyleMedium2" ' Professional Blue Theme

MsgBox "Data Cleaning and Table Formatting Complete!", vbInformation,
"Success"

End Sub

